

SOC Analyst Lab Report

SYN Flood DoS Attack — Simulation & Analysis

June 2026 | Kali Linux VM | Tools: hping3, Nmap, Wireshark

1. Lab Overview

This lab demonstrates a TCP SYN flood Denial of Service (DoS) attack in a fully controlled, isolated environment. The objective was to simulate the attack, capture the resulting network traffic, analyse the packet-level indicators, and apply SOC detection and response methodology. No external or production systems were targeted at any time.

2. Lab Environment

Field	Details
Operating System	Kali Linux (Virtual Machine)
Attack Tool	hping3 v3.a2.ds2-11-kali1
Reconnaissance Tool	Nmap 7.99
Capture Tool	Wireshark 4.6.6
Network Interface	eth0 / any
Source IP	[redacted] / 10.0.0.4
Destination IP	[redacted].95.254
Target Port	TCP Port 80 (HTTP)
Attack Type	TCP SYN Flood (DoS)

3. Tool Installation & Verification

All tools were installed using the Kali Linux APT package manager. The following screenshots document each installation step.

3.1 Installing hping3

```
sudo apt install hping3 -y
```


3.4 Verifying Wireshark Version

wireshark --version

```
File Edit Tabs Help
Setting up libqt6waylandcompositor6:amd64 (6.10.2-5) ...
Setting up qt6-wayland:amd64 (6.10.2-5) ...
Processing triggers for ncolor-icon-theme (0.18-2) ...
Processing triggers for libc-bin (2.42-5) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for shared-mime-info (2.4-5+b3) ...
Processing triggers for desktop-file-utils (0.28-1) ...

--(base) [~]
$ wireshark --version
Wireshark 4.6.6.

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This is free software; see the file named COPYING in the distribution. There is
NO WARRANTY, not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compile-time info:
  Bit width: 64-bit
  Compiler: GCC 15.2.0
  GLib: 2.88.1
  With:
    +brotli
    +Gcrypt 1.12.2
    +GnuTLS 3.8.13 and PKCS#11
    +Kerberos (MIT)
    +libnl 3
    +libpcap
    +libsm 0.4.8
    +libxml2 2.15.2
    +Lua 5.4.8
    +LZ4 1.10.0
    +MaxMind
    +Minizip 1.3.2
    +nghttp2 1.69.0
    +nghttp3 1.15.0
    +PCRE2 10.46 2025-08-27
    +POSIX capabilities (Linux)
    +Qt 6.10.2
    +QtDBus
    +QtMultimedia
    +Snappy 1.2.2
    +xxhash 0.8.3
    +zlib 1.3.2
    +Zstandard 1.5.7
```

Figure 4: Wireshark 4.6.6 confirmed installed successfully

4. Reconnaissance — Nmap Scan

Prior to running the attack simulation, an Nmap scan was performed against the target IP to identify open ports and understand the host's network posture.

nmap [target IP]

```
$ nmap 95.254
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-13 16:29 EDT
Nmap scan report for 95.254
Host is up (0.0048s latency).
All 1000 scanned ports on 95.254 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 17.38 seconds
```

Figure 5: Nmap scan result — all 1000 ports filtered (no-response)

The scan returned all 1000 TCP ports in a filtered state with no response, indicating a firewall or security device (such as a WAF or CDN edge) was silently dropping packets. This is consistent with Cloudflare-protected infrastructure.

5. Attack Simulation — SYN Flood

5.1 Starting Wireshark Capture

Wireshark was launched with elevated privileges and configured to capture on the loopback interface before the attack was initiated.

```
sudo wireshark
```

```
--(~/warsaw@kali)-[~]
└─$ sudo wireshark
** (wireshark:6549) 21:59:55.296704 [GUI WARNING] -- Session Dbus not running.
** (wireshark:6549) 21:59:55.296814 [GUI WARNING] -- Application will not react to setting changes.
Check your Dbus installation.
** (wireshark:6549) 21:59:55.296956 [GUI WARNING] -- dbus connection failed. Last error: QDBusError("org.freedesktop.DBus.Error.Spawn.ExecFailed", "/usr/bin/dbus-launch terminated abnor
ally without any error message")
** (wireshark:6549) 21:59:55.353897 [GUI WARNING] -- Session Dbus not running.
** (wireshark:6549) 21:59:55.354002 [GUI WARNING] -- Application will not react to setting changes.
Check your Dbus installation.
** (wireshark:6549) 21:59:55.354065 [GUI WARNING] -- dbus connection failed. Last error: QDBusError("org.freedesktop.DBus.Error.Spawn.ExecFailed", "/usr/bin/dbus-launch terminated abnor
ally without any error message")
** (wireshark:6549) 21:59:58.346458 [Capture MESSAGE] -- Capture Start ...
** (wireshark:6549) 21:59:58.396451 [Capture MESSAGE] -- Capture started
** (wireshark:6549) 21:59:58.396524 [Capture MESSAGE] -- File: "/tmp/wireshark-LoopbackMYJN03.pcapng"
```

Figure 6: Wireshark launched via sudo — capture started on loopback interface

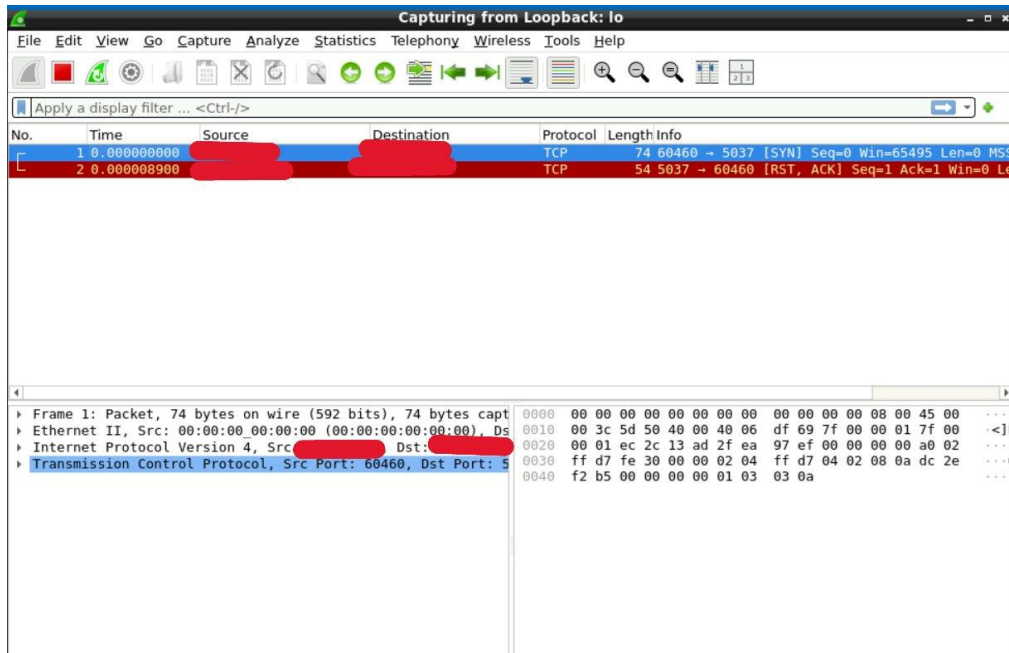


Figure 7: Wireshark capturing on loopback (lo) interface — initial packets visible

5.2 Executing the SYN Flood

The hping3 SYN flood was executed from a second terminal while Wireshark continued capturing traffic in the background.

```
sudo hping3 -S --flood -V -p 80 [target IP]
```

Flag	Description
-S	Set TCP SYN flag only — no handshake completion
--flood	Send packets as fast as possible
-V	Verbose output
-p 80	Target TCP port 80 (HTTP)
[target IP]	Destination host for the flood

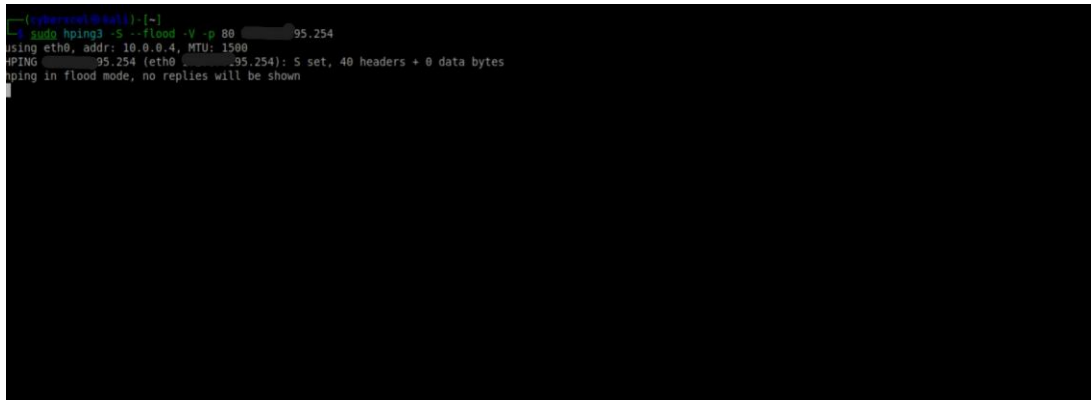


Figure 8: hping3 SYN flood running — flood mode active on eth0

hping3 confirmed flood mode was active, transmitting 3,486,590 packets with 0 replies received and 100% packet loss — consistent with a firewall dropping all inbound SYN packets at the target.

6. Wireshark Traffic Analysis

After stopping the hping3 flood, a Wireshark display filter was applied to isolate only TCP SYN packets with no ACK flag — the definitive signature of a SYN flood attack.

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

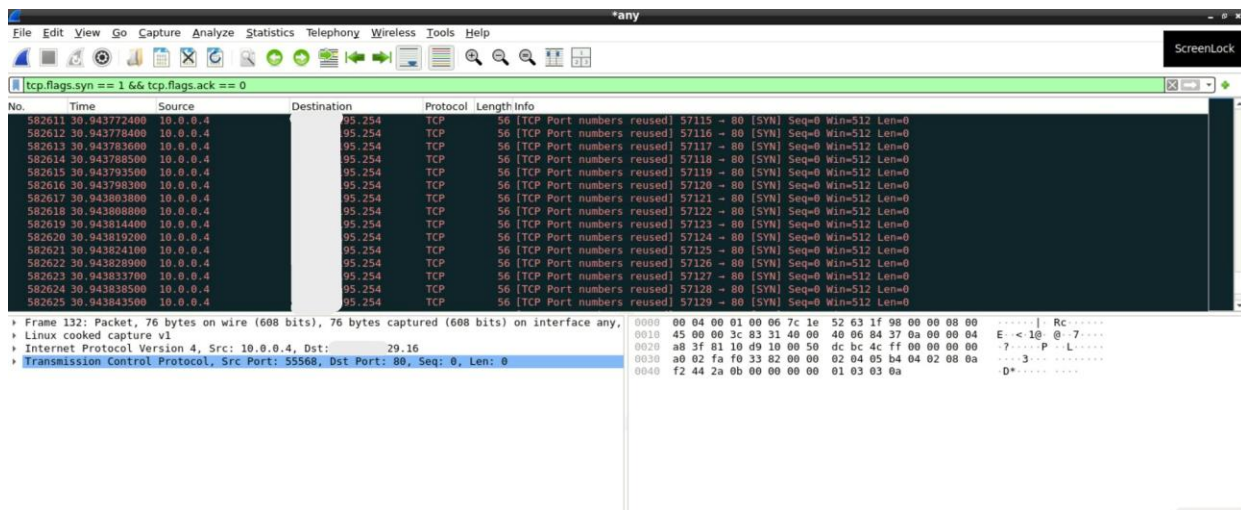


Figure 9: Wireshark filtered view — 582,625+ TCP SYN packets captured

Key Observations

- 582,625+ TCP SYN packets captured within the capture window
- All packets originated from source IP 10.0.0.4 targeting the destination on port 80

- No SYN-ACK responses observed — the three-way handshake was never completed
- Packet timestamps are microseconds apart, confirming flood-rate transmission
- Win=512 and Len=0 are characteristic hping3 SYN flood packet signatures
- TCP port numbers reused across packets — typical of rapid port cycling in flood tools
- Packet size consistent at 56 bytes throughout the capture

7. SOC Detection Perspective

From a SOC analyst standpoint, the following indicators in a real-world environment would trigger detection and escalation:

Detection Indicators

- Sudden spike in inbound TCP SYN packets to a single destination port
- High volume of half-open connections not progressing beyond SYN
- Single source IP generating abnormally high connection attempts per second
- Service degradation or unavailability reported on the targeted port
- IDS/IPS alerts for SYN flood or connection rate threshold breach
- SIEM alert triggered by connection rate anomaly analytics rules

MITRE ATT&CK Mapping

Field	Details
Tactic	Impact
Technique	T1498 — Network Denial of Service
Sub-technique	T1498.001 — Direct Network Flood
Protocol Exploited	TCP Three-Way Handshake

8. Mitigation Strategies

- Enable SYN cookies on target servers to handle incomplete handshakes without allocating state
- Configure rate limiting on the firewall to restrict SYN packets per source IP
- Deploy IDS/IPS with SYN flood detection signatures (e.g. Snort, Suricata)
- Use a WAF or CDN service such as Cloudflare to absorb and filter volumetric attacks
- Implement geo-blocking or IP reputation filtering for known malicious sources
- Create SIEM analytics rules to alert on anomalous SYN packet rates
- Use blackhole routing (RTBH) at ISP level for large-scale DDoS mitigation

9. Lessons Learned

- hping3 is highly effective for simulating DoS traffic and understanding packet-level attack mechanics
- Wireshark display filters are essential for isolating attack traffic from background noise
- SYN floods are easily identifiable in packet captures due to the absence of SYN-ACK responses
- Understanding the TCP three-way handshake is fundamental to recognising and responding to this attack class
- Proper lab setup using isolated VM environments is critical to ensure ethical, controlled experimentation
- Real-world SYN floods are typically distributed (DDoS), making single-IP blocking insufficient — layered defences are required